

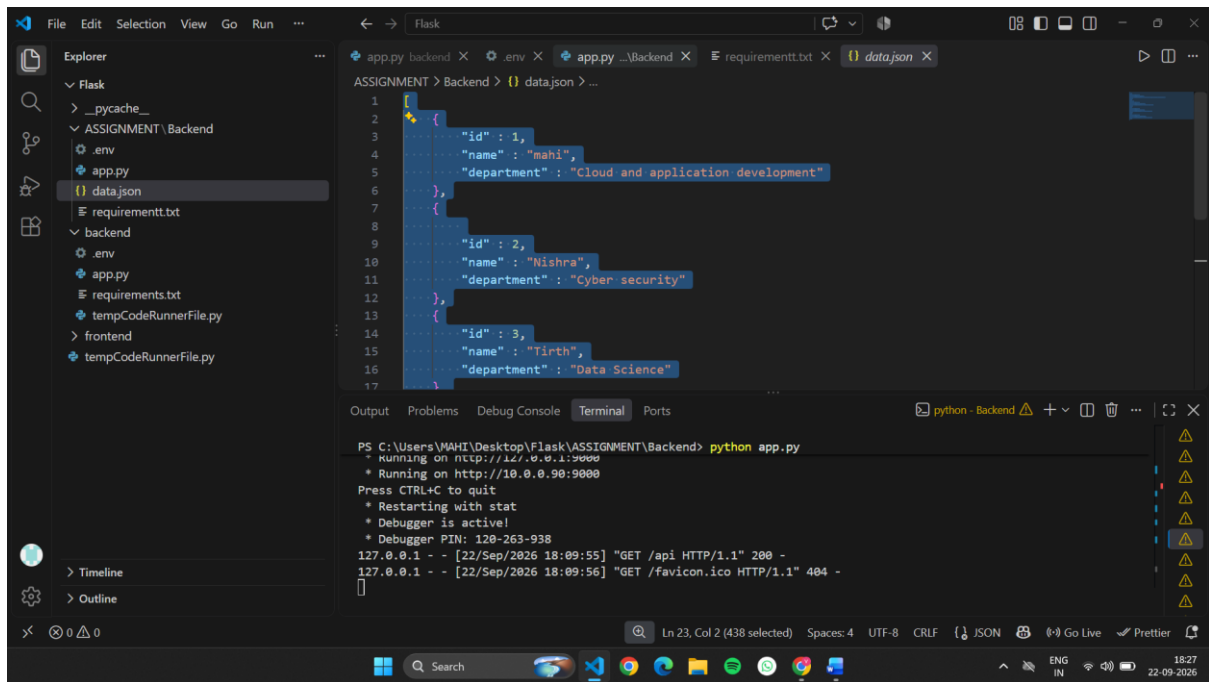
NAME : MAHI

Flask and MongoDB

TASK 1:

Create a backend folder inside that create a file data.json. This file stores the data that will be read by the Flask application.

```
[
  {
    "id" : 1,
    "name" : "mahi",
    "department" : "Cloud and application development"
  },
  {
    "id" : 2,
    "name" : "Nishra",
    "department" : "Cyber security"
  },
  {
    "id" : 3,
    "name" : "Tirth",
    "department" : "Data Science"
  },
  {
    "id" : 4,
    "name" : "Nishant",
    "department" : "Data Science"
  }
]
```



Next create a file which name is app.py. this file return the data of data.json

```
@app.route("/api")
```

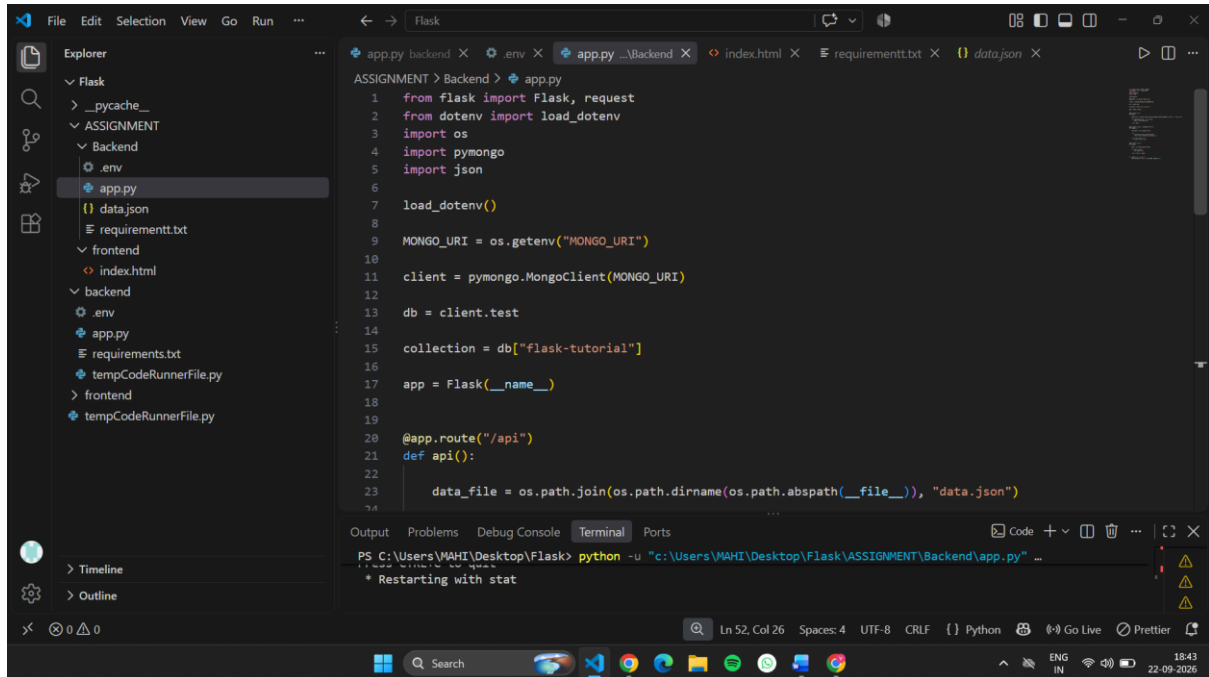
```
def api():
```

```
    data_file = os.path.join(os.path.dirname(os.path.abspath(__file__)), "data.json")
```

```
    with open(data_file, "r") as file:
```

```
        data = json.load(file)
```

return data



```
1 from flask import Flask, request
2 from dotenv import load_dotenv
3 import os
4 import pymongo
5 import json
6
7 load_dotenv()
8
9 MONGO_URI = os.getenv("MONGO_URI")
10
11 client = pymongo.MongoClient(MONGO_URI)
12
13 db = client.test
14
15 collection = db["flask-tutorial"]
16
17 app = Flask(__name__)
18
19
20 @app.route("/api")
21 def api():
22
23     data_file = os.path.join(os.path.dirname(os.path.abspath(__file__)), "data.json")
24
```

PS C:\Users\MAHI\Desktop\Flask> python -u "c:\Users\MAHI\Desktop\Flask\ASSIGNMENT\Backend\app.py" ...

* Restarting with stat

INSTALL FLASK:

Pip install flask

Run the Flask Application

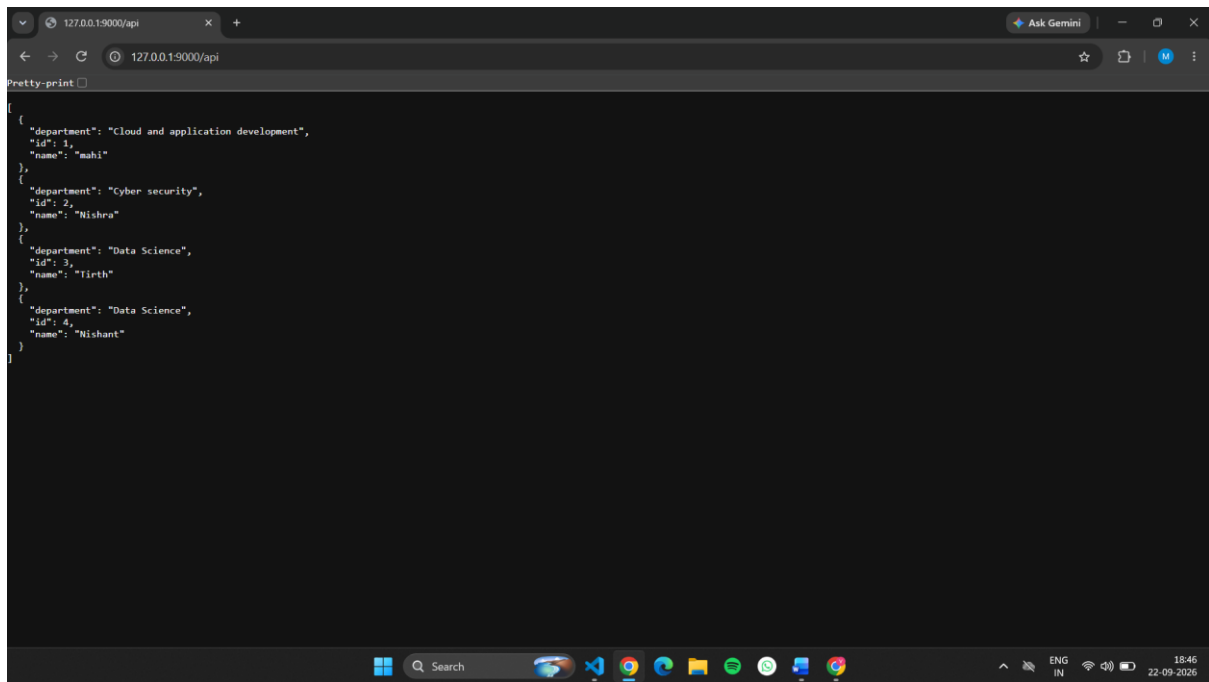
Run the following command in the project folder:

python app.py

After starting the application, <http://127.0.0.1:9000>

Then I open URL like : <http://127.0.0.1:9000/api>

And that's the final output of task 1:



TASK 2: Frontend Form with MongoDB Atlas

We create form for insert data and message success and error.

We create app.py

```
from flask import Flask, render_template, request
```

```
from pymongo import MongoClient
```

```
from dotenv import load_dotenv
```

```
import os
```

```
load_dotenv()
```

```
app = Flask(__name__, template_folder='../frontend')
```

```
client = MongoClient("mongodb://localhost:27017/")
```

```
db = client["student_db"]
```

```
collection = db["students"]
```

```
@app.route("/")
```

```
def home():  
    return render_template("index.html")  
  
@app.route("/submit", methods=["POST"])  
def submit():  
  
    try:  
        name = request.form["name"]  
        email = request.form["email"]  
        age = request.form["age"]  
  
        raise Exception("This is a test error")  
  
        student = {  
            "name": name,  
            "email": email,  
            "age": int(age)  
        }  
  
        collection.insert_one(student)  
  
        return render_template("success.html")  
  
    except Exception as e:  
  
        return render_template(  
            "index.html",  
            error=str(e)  
        )  
  
if __name__ == "__main__":
```

```
app.run(port=9000, debug=True)
```

than we create frontend form index.html inside frontend folder.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Student Form</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Student Registration Form</h1>
```

```
  <!-- {% if error %}
```

```
    <p style="color: red;">
```

```
      Error: {{ error }}
```

```
    </p>
```

```
{% endif %} -->
```

```
<form action="/submit" method="POST">
```

```
  <label>Name:</label>
```

```
  <input type="text" name="name" required>
```

```
  <br><br>
```

```
  <label>Email:</label>
```

```
  <input type="email" name="email" required>
```

```
  <br><br>
```

```
  <label>Age:</label>
```

```
  <input type="number" name="age" required>
```

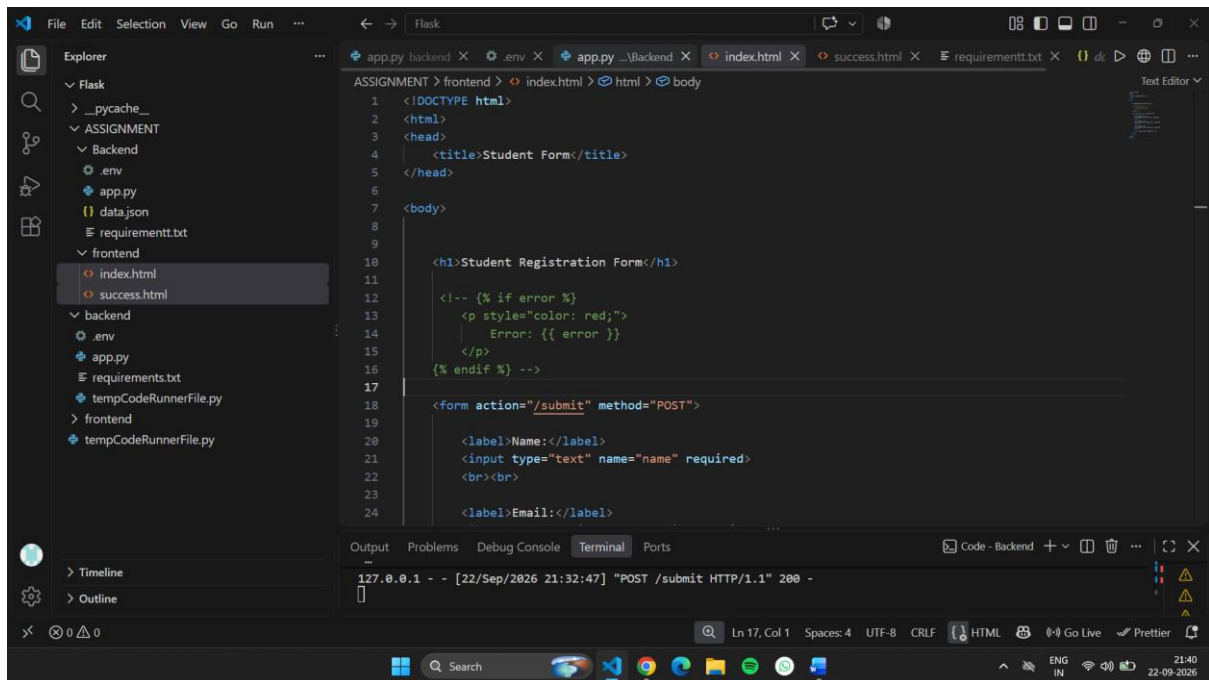
```
  <br><br>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</body>
```

```
</html>
```



Then we create success.html also in frontend folder to show form submitted successful.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Success</title>
```

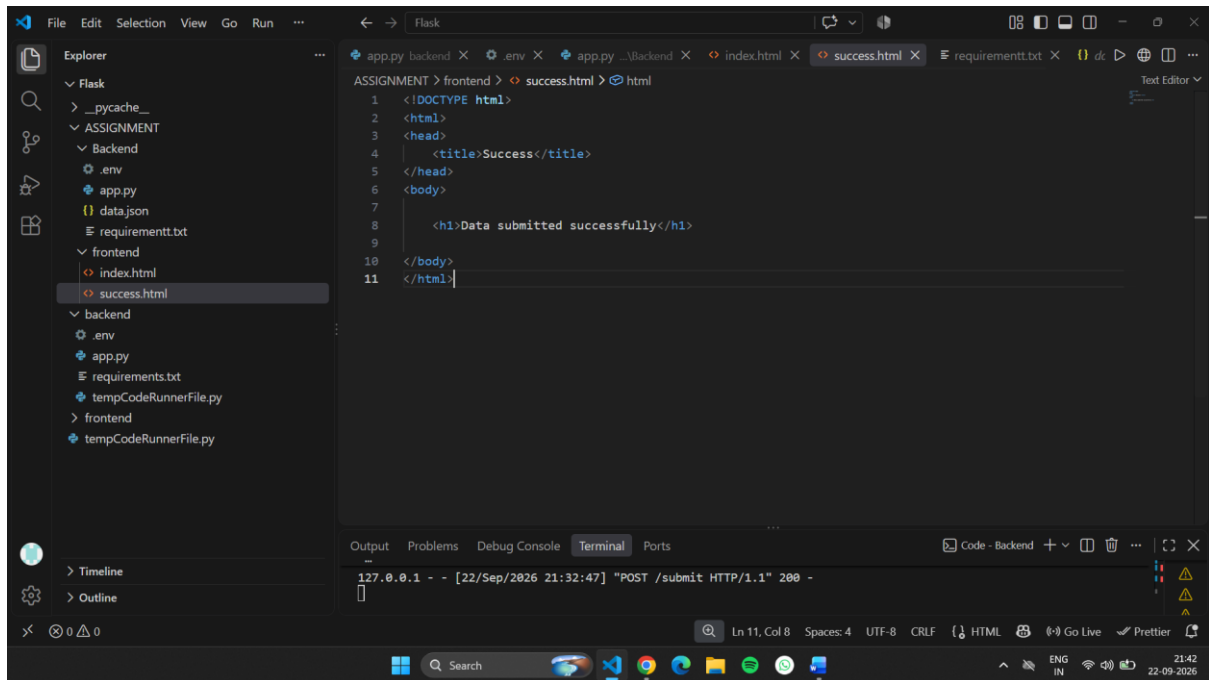
```
</head>
```

```
<body>
```

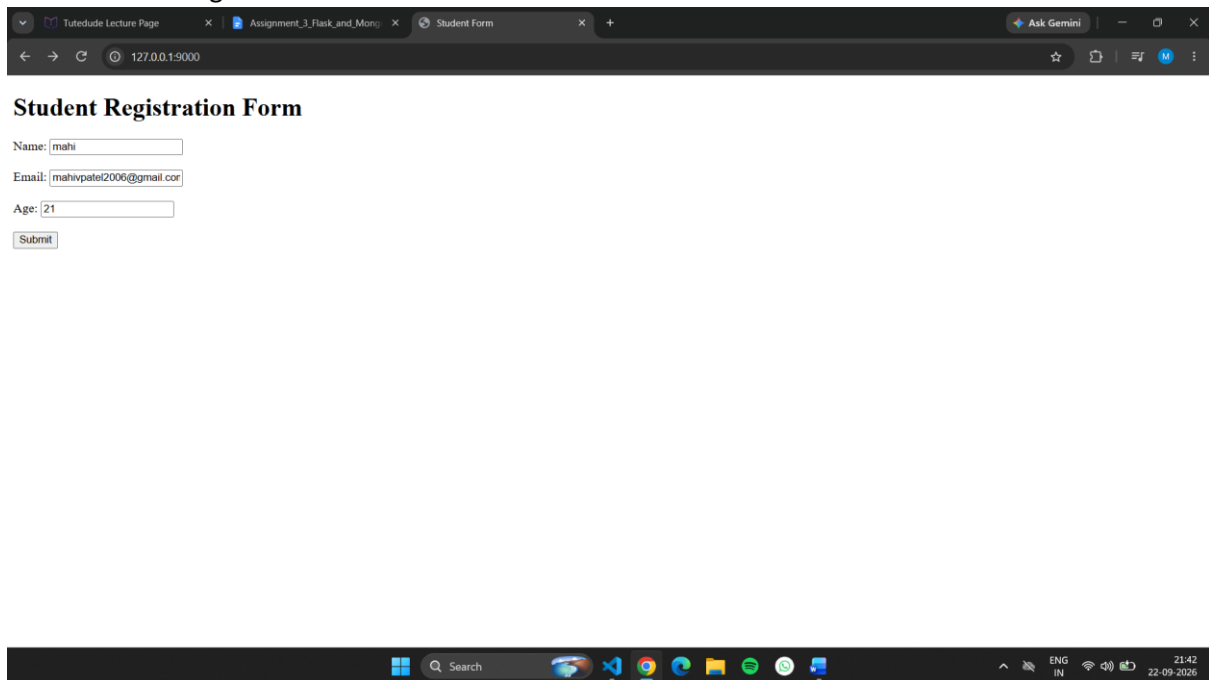
```
<h1>Data submitted successfully</h1>
```

</body>

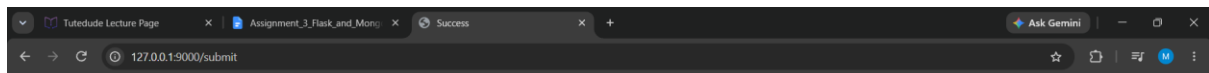
</html>



This is student registration form



Than I click on submit that show to data submit successfully



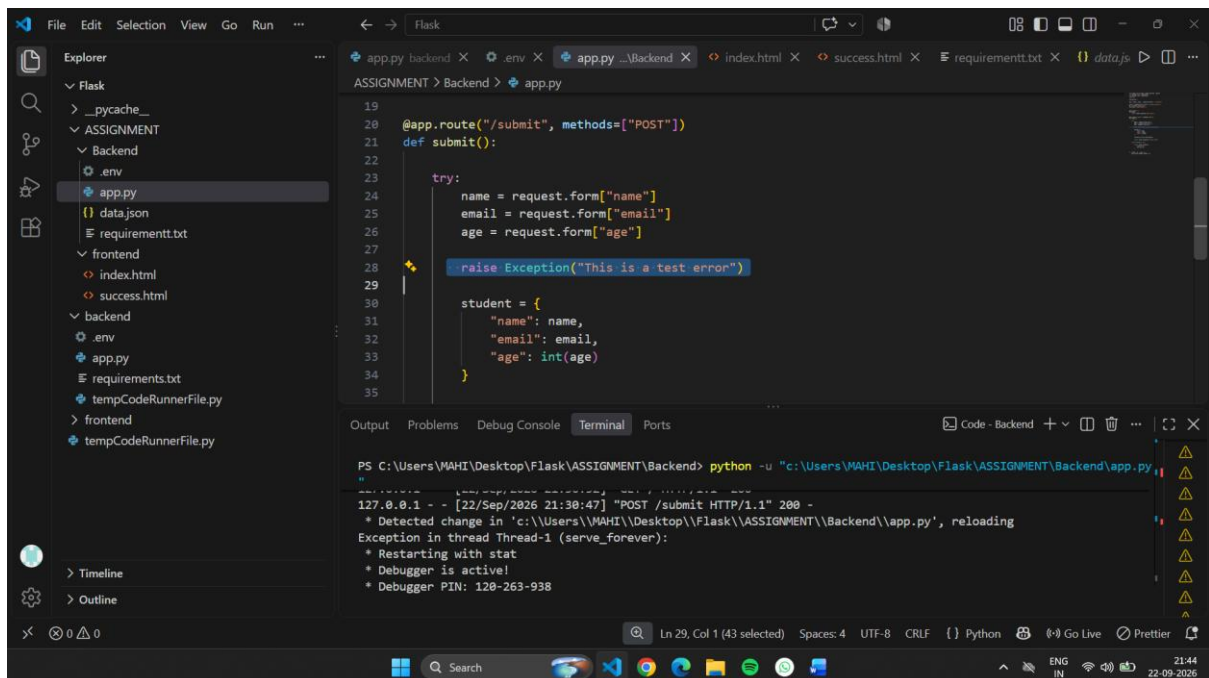
Data submitted successfully



Than for error message

In app.py add this line:

```
raise Exception("This is a test error")
```



And index.html we add this line:

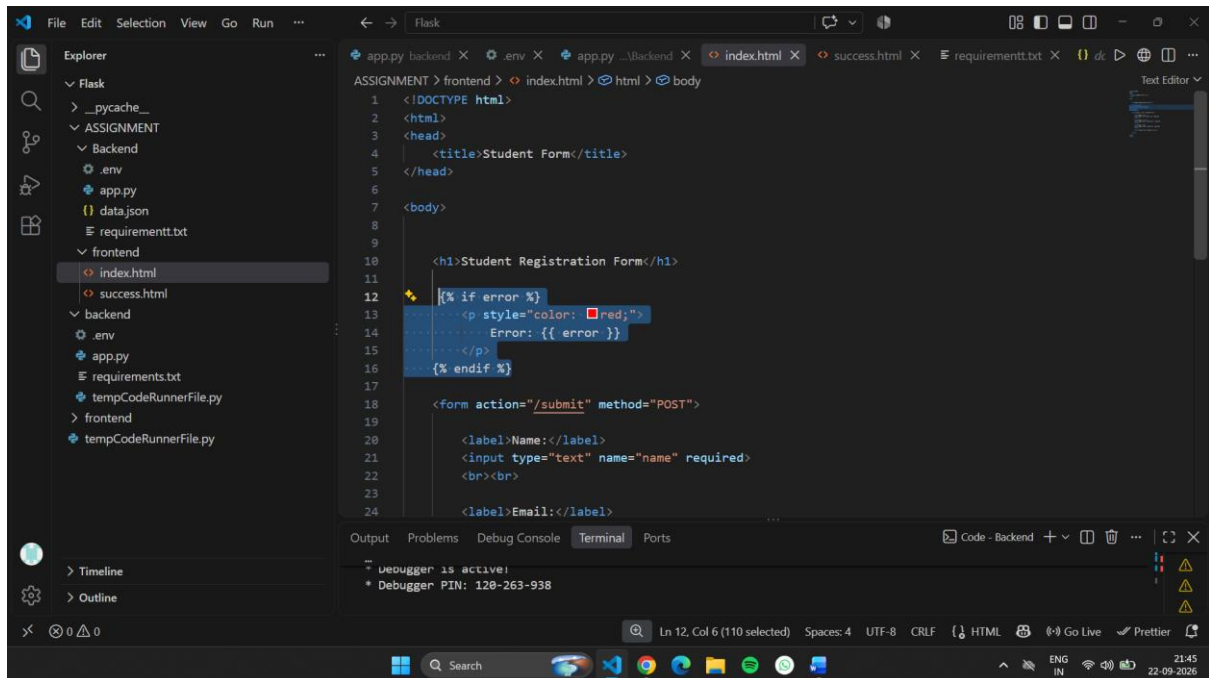
```
{% if error %}
```

```
<p style="color: red;">
```

```
Error: {{ error }}
```

</p>

{% endif %}



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Student Form</title>
5 </head>
6
7 <body>
8
9   <h1>Student Registration Form</h1>
10
11   {% if error %}
12     <p style="color: red;">
13       Error: {{ error }}
14     </p>
15   {% endif %}
16
17   <form action="/submit" method="POST">
18
19     <label>Name:</label>
20     <input type="text" name="name" required>
21     <br><br>
22
23     <label>Email:</label>
```

this lines use for error message and also we use p style for adding color

And that's the output of error checking:

